Where the log $\frac{\vee}{k^+}$ of Group III comes from: step by step, with your Row VI

Rows: your `Other_terms.tex`. In your `BFKL.tex`, Group III Rows III, IV, V are "NA". The rows meant here are the rows of `Other_terms.tex` whose line 2 has $\log(\frac{\vee}{k^+}-1)$: Group III Row III (both parts), Row V and Row VI (Row IV has only four ρ's), and Group II Row IV. Notation as in your files.

1 The answer in short

Your Row VI is correct up to and including the plus prescription. After it, the only logarithm you can *see* is $\big[\log\frac{k^+}{\Lambda}+\log(\frac{\vee}{k^+}-1)\big]$ in line 2. But line 1 is not independent of $\vee$: its upper limit is $\vee-k^+$. There are two different $\log\frac{\vee}{k^+}$ in the row:

1. **The visible one**, $\log(\frac{\vee}{k^+}-1)=\log\frac{\vee}{k^+}+\log(1-\frac{k^+}{\vee})$ in line 2. It is *not* a BFKL term, and you are right to leave it out of `BFKL.tex`. It is cancelled exactly by the subtraction $-f(0)$ of $\left[\frac{1}{p^+}\right]_+$ in line 1, integrated up to $p^+=\vee-k^+$ (Step 5).

2. **A hidden one**, in line 1, in the term $-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}$ of the bracket. As written it carries no logarithm, but its p^+ integral grows like $\log\frac{\vee}{k^+}$ for $\vee\to\infty$. This shows up only once you look at what the y' integral does at large p^+ (Steps 3, 4). This one *is* the BFKL term; it is called (B) (Step 6).

So, row by row: Row VI $=\log\frac{k^+}{\Lambda}\times$(JIMWLK part) + (B) + a rest that is finite for $\vee\to\infty$ (Step 7). Without (B), the BFKL list misses the terms in which the measured gluon is emitted by the gluon at z: 9 of the 67 colour/kernel structures of the two-ρ sum then differ from BFKL⊗LO (Sec. 11). Without $-$(B), the "rest" still grows like $\log\frac{\vee}{k^+}$.

2 Step 1: your Row VI

From `Other_terms.tex` (your last equation for Row VI):

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = {}& \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\log\Big(\frac{\vee}{k^+}-1\Big)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big]
\end{aligned}
\tag{2.1}
$$

Call the colour factor

$$
\mathcal{C}(y')=\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big]. \tag{2.2}
$$

Only the first bracket depends on y', through $U^{e'c'}(y')$; $\mathcal{C}(z)$ is $\mathcal{C}(y')$ with $U^{e'c'}(y')\to U^{e'c'}(z)$.

3 Step 2: line 1 is three terms

Contract the Kronecker deltas of the three terms of the bracket of line 1 and name the transverse integrals:

$$
\begin{aligned}
F_a(p^+) &= \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)\cdot(x'-y')\,(y-w)\cdot(x-z)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\,\mathcal{C}(y') && (\delta_{k'm}\delta_{ij}),\\
F_b(p^+) &= \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\,\mathcal{C}(y') && (\delta_{jm}\delta_{ik'}),\\
F_c(p^+) &= \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)\cdot(y-w)\,(x'-y')\cdot(x-z)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\,\mathcal{C}(y') && (\delta_{im}\delta_{jk'}).
\end{aligned}
\tag{3.1}
$$

With $C=\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}$ and your $\left[\frac{1}{p^+}\right]_+ f(p^+)=\frac{1}{p^+}\big(f(p^+)-f(0)\big)$, Eq. (2.1) is

$$
\text{line 1}=C\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\left[\frac{F_a(p^+)}{p^++k^+}-\frac{F_b(p^+)-F_b(0)}{p^+}-\frac{F_c(p^+)}{k^+}\right], \tag{3.2}
$$

$$\text{line 2} = -\frac{C}{2\pi}\,F_b(0)\left[\log\Big(\frac{k^+}{\Lambda}\Big)+\log\Big(\frac{\vee}{k^+}-1\Big)\right], \qquad \int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}=\log\frac{k^+}{\Lambda}+\log\Big(\frac{\vee}{k^+}-1\Big). \tag{3.3}$$

$F_b(0)$ has no p^+ phase: it is the integral of your line 2. ($\frac{C}{2\pi}=\frac{1}{(2\pi)^3}\frac{ig^4 f^{c'd'a}}{8\pi^5}\frac{1}{k^+}$, the prefactor of your line 2.)

4 Step 3: what the y' integral does at large p^+

The two phases combine to one:

$$e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}=e^{-ik\cdot(z-w)}\;e^{-i\,t\,k\cdot(y'-z)}, \qquad t=\frac{p^++k^+}{k^+}, \quad 1\le t\le\frac{\vee}{k^+}. \tag{4.1}$$

So the y' integral is a Fourier transform at the momentum $q=t\,k$, which grows with p^+. All three integrands have the factor $\frac{(y'-z)^m}{(y'-z)^2}$, singular at $y'=z$; the rest, $G(y')$, is smooth there. The exact transform

$$\int d^2r\;e^{-iq\cdot r}\,\frac{r^m}{r^2}=-2\pi i\,\frac{q^m}{q^2} \qquad \Big(\tfrac{r^m}{r^2}=\partial^m\tfrac{1}{2}\log r^2,\ \int d^2r\,e^{-iq\cdot r}\log r^2=-\tfrac{4\pi}{q^2}\Big) \tag{4.2}$$

decreases only like $\frac{1}{q}$. For large q the oscillating phase cancels everything except $|y'-z|\lesssim\frac{1}{|q|}$, where $G(y')\simeq G(z)$:

$$\int_{y'} e^{-iq\cdot(y'-z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,G(y')=-2\pi i\,\frac{q^m}{q^2}\,G(z)+O\Big(\frac{1}{q^2}\Big). \tag{4.3}$$

With $\frac{q^m}{q^2}=\frac{1}{t}\frac{k^m}{k^2}=\frac{k^+}{p^++k^+}\frac{k^m}{k^2}$, and $\int_{y'}e^{-ik\cdot(y'-w)}\frac{(y'-z)^m}{(y'-z)^2}=e^{-ik\cdot(z-w)}\big(-2\pi i\frac{k^m}{k^2}\big)$, this says

$$\boxed{F_n(p^+)=\frac{k^+}{p^++k^+}\,F_n^z+O\Big(\frac{k^{+2}}{p^{+2}}\Big)}, \qquad n=a,b,c, \tag{4.4}$$

where F_n^z is F_n at $p^+=0$ with $y'\to z$ everywhere except in $\frac{(y'-z)^m}{(y'-z)^2}$ and in $e^{-ik\cdot(y'-w)}$. For the third term:

$$F_c^z=\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\;\mathcal{C}(z). \tag{4.5}$$

- In words: the w' integral set $w'=\frac{(p^++k^+)y'-p^+z}{k^+}$, i.e. $y'=\frac{k^+w'+p^+z}{p^++k^+}$. For $p^+\gg k^+$ the parent sits on z: the measured gluon (k^+) is emitted by the much faster gluon at z. This is the evolution of the projectile between k^+ and $\vee$, i.e. BFKL.
- The second singular point, $y'\to x'$, also gives a $\frac{k^+}{p^++k^+}$ tail, but with the phase $e^{-i\frac{p^+}{k^+}k\cdot(x'-z)}$, which oscillates in p^+. Its p^+ integral converges, so it gives no logarithm.
- Eq. (4.3) is exact for a Gaussian G (Sec. 9), and was checked numerically for a non-Gaussian G in `BFKL_final.pdf` (ratio 0.999996, 1.000015, 1.000054, 1.000205 at $q=6k,11k,21k,41k$).

5 Step 4: which term of line 1 can grow like $\log\frac{\vee}{k^+}$

Put Eq. (4.4) into Eq. (3.2). A p^+ integrand that falls like $\frac{1}{p^{+2}}$ gives a finite integral for $\vee\to\infty$. One that falls like $\frac{1}{p^+}$ gives $\log\frac{\vee}{k^+}$.

term of line 1	p^+ factor	integrand at $p^+\gg k^+$	$\int^{\vee-k^+}dp^+$
$\delta_{k'm}\delta_{ij}$	$\frac{1}{p^++k^+}$	$\frac{k^+F_a^z}{(p^++k^+)^2}$	finite
$\delta_{jm}\delta_{ik'}$, the $f(p^+)$ of $[\frac{1}{p^+}]_+$	$-\frac{1}{p^+}$	$-\frac{k^+F_b^z}{p^+(p^++k^+)}$	finite
$\delta_{jm}\delta_{ik'}$, the $-f(0)$ of $[\frac{1}{p^+}]_+$	$-\frac{1}{p^+}$	$+\frac{F_b(0)}{p^+}$	$+F_b(0)\log(\frac{\vee}{k^+}-1)+\ldots$ (Step 5)
$\delta_{im}\delta_{jk'}$	$-\frac{1}{k^+}$	$-\frac{F_c^z}{p^++k^+}$	$-F_c^z\log\frac{\vee}{k^+}+\ldots$ (Step 6)

So two pieces of line 1 grow like $\log\frac{\vee}{k^+}$: the subtraction of the plus prescription, and the $-\frac{1}{k^+}$ term. They have different coefficients, $F_b(0)$ and F_c^z, and they do different things.

6 Step 5: the $\log(\frac{\vee}{k^+}-1)$ of line 2 is cancelled

Split the plus-prescribed integral at $p^+=k^+$ (for $\vee>2k^+$) and use $\int_{k^+}^{\vee-k^+}\frac{dp^+}{p^+}=\log(\frac{\vee}{k^+}-1)$, exactly:

$$-\int_0^{\vee-k^+}dp^+\,\frac{F_b(p^+)-F_b(0)}{p^+}=-\int_0^{k^+}dp^+\,\frac{F_b(p^+)-F_b(0)}{p^+}-\int_{k^+}^{\vee-k^+}dp^+\,\frac{F_b(p^+)}{p^+}+F_b(0)\log\Big(\frac{\vee}{k^+}-1\Big). \tag{6.1}$$

Add line 2, Eq. (3.3): $\log(\frac{\vee}{k^+}-1)$ drops out,

$$\text{line 1}\big|_{\delta_{jm}\delta_{ik'}} + \text{line 2} = \frac{C}{2\pi}\Big[-F_b(0)\log\frac{k^+}{\Lambda} - \int_0^{k^+} dp^+\,\frac{F_b(p^+)-F_b(0)}{p^+} - \int_{k^+}^{\vee-k^+} dp^+\,\frac{F_b(p^+)}{p^+}\Big]. \tag{6.2}$$

The two integrals are finite for $\vee\to\infty$ ($F_b(p^+)\sim\frac{k^+}{p^+}$). What is left is JIMWLK ($\log\frac{k^+}{\Lambda}$) and a finite rest, with **no** $\log\frac{\vee}{k^+}$. Your $\log(\frac{\vee}{k^+}-1)$ only appears because $\big[\frac{1}{p^+}\big]_+$ subtracts $F_b(0)$ all the way up to $p^+=\vee-k^+$; the subtraction and line 2 cancel each other.

7 Step 6: the hidden BFKL term (B) in the $-\frac{1}{k^+}$ term

Since $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+}=\log\frac{\vee}{k^+}$ exactly,

$$-\frac{1}{k^+}\int_0^{\vee-k^+} dp^+\,F_c(p^+) = -F_c^z\,\log\frac{\vee}{k^+} \;-\; \frac{1}{k^+}\int_0^{\vee-k^+} dp^+\Big[F_c(p^+)-\frac{k^+}{p^++k^+}\,F_c^z\Big], \tag{7.1}$$

and by Eq. (4.4) the last integrand falls like $\frac{1}{p^{+2}}$, so the integral is finite for $\vee\to\infty$. The first term, times $\frac{C}{2\pi}$, is the BFKL term of Row VI. With Eq. (4.5) (the contractions of $\delta_{im}\delta_{jk'}$: $(y'-z)\cdot(y-w)$ and $(x'-y')\cdot(x-z)\to(x'-z)\cdot(x-z)$; $U^{e'c'}(y')\to U^{e'c'}(z)$):

$$\begin{aligned}(\mathrm{B})_{\mathrm{VI}} = &-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\ &\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\ &\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big]\end{aligned} \tag{7.2}$$

This is the term (B) of Group III Row VI in `BFKL_rows.pdf`, Eq. (4.8).

8 Step 7: Row VI = JIMWLK + BFKL + rest

Collecting Steps 5 and 6 (the $\delta_{k'm}\delta_{ij}$ term is finite):

$$\text{Row VI} = \underbrace{-\frac{C}{2\pi}\,F_b(0)\,\log\frac{k^+}{\Lambda}}_{\text{JIMWLK (the }\log\frac{k^+}{\Lambda}\text{ of your line 2)}} + \underbrace{(\mathrm{B})_{\mathrm{VI}}}_{\text{BFKL}} + \mathrm{Rest}_{\mathrm{VI}}. \tag{8.1}$$

The rest can be written in two equivalent ways:

- **In your form** (as in `Rest_terms.pdf`): line 1 as you have it, + line 2 with $\log(\frac{\vee}{k^+}-1)$ only, −(B):

$$\mathrm{Rest}_{\mathrm{VI}} = \text{line 1} - \frac{C}{2\pi}\,F_b(0)\log\Big(\frac{\vee}{k^+}-1\Big) + C\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{F_c^z}{p^++k^+}. \tag{8.2}$$

  The last term is −(B), written with $\log\frac{\vee}{k^+}=\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+}$. It is a subtraction at the hard end $p^+\to\vee$, as $[\frac{1}{p^+}]_+$ is one at the soft end $p^+\to 0$.

- **Manifestly finite** (Eqs. (6.2), (7.1)): every integrand falls like $\frac{1}{p^{+2}}$, so one may even set $\vee\to\infty$:

$$\mathrm{Rest}_{\mathrm{VI}} = \frac{C}{2\pi}\Big\{\int_0^{\vee-k^+} dp^+\,\frac{F_a(p^+)}{p^++k^+} - \int_0^{k^+} dp^+\,\frac{F_b(p^+)-F_b(0)}{p^+} - \int_{k^+}^{\vee-k^+} dp^+\,\frac{F_b(p^+)}{p^+} - \frac{1}{k^+}\int_0^{\vee-k^+} dp^+\Big[F_c(p^+)-\frac{k^+F_c^z}{p^++k^+}\Big]\Big\}. \tag{8.3}$$

9 Step 8: numbers (toy model)

To see Steps 4–6 with numbers, take $|k|=1$ and replace, in the y' integral, everything except $\frac{(y'-z)^m}{(y'-z)^2}$ and the phase by $e^{-(y'-z)^2/2}$. Then Eq. (4.3) is exact:

$$\int d^2y'\,e^{-i\,t\,k\cdot(y'-z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,e^{-(y'-z)^2/2} = -2\pi i\,\frac{k^m}{k^2}\,\frac{1-e^{-t^2/2}}{t}, \tag{9.1}$$

so $F(p^+)=F^z\,f(t)$ with $f(t)=\frac{1-e^{-t^2/2}}{t}\to\frac{1}{t}=\frac{k^+}{p^++k^+}$, $f(1)=F(0)/F^z=0.3935$. Use the same F in all three terms (only the p^+ factor differs). In units of $\frac{C}{2\pi}F^z$ (`why_toy.py`):

$\frac{\vee}{k^+}$	$\delta_{k'm}\delta_{ij}$	$[\frac{1}{p^+}]_+$ term	line 2: $\log(\frac{\vee}{k^+}-1)$	sum of the two	$-\frac{1}{k^+}$ term	$-\frac{1}{k^+}$ term $+\log\frac{\vee}{k^+}$
10^1	$+0.69116$	$+0.17814$	-0.86454	-0.68640	-2.02270	$+0.27989$
10^2	$+0.78116$	$+1.02633$	-1.80804	-0.78171	-4.32528	$+0.27989$
10^3	$+0.79016$	$+1.92683$	-2.71760	-0.79076	-6.62787	$+0.27989$
10^4	$+0.79106$	$+2.83228$	-3.62395	-0.79166	-8.93045	$+0.27989$
10^6	$+0.79116$	$+4.64422$	-5.43598	-0.79176	-13.53562	$+0.27989$

- The $\delta_{k'm}\delta_{ij}$ term converges.
- The $[\frac{1}{p^+}]_+$ term grows by $f(1)\log 10 = 0.9060$ per decade, and line 2 falls by the same amount; their sum converges (Step 5).
- The $-\frac{1}{k^+}$ term falls by $\log 10 = 2.3026$ per decade. It is $-\log\frac{\vee}{k^+}$ plus a constant, exactly: the coefficient of the logarithm is F^z, the $y'\to z$ value (Step 6). Adding $+\log\frac{\vee}{k^+}$, i.e. subtracting (B), leaves a constant.

10 The other rows

Group III Row III, first part. The same three terms, with $\frac{ig^4 f^{c'd'a}}{8\pi^4}$ and the colour factor with the anticommutators. Steps 2–7 are unchanged:

$$\begin{aligned}(\mathrm{B})_{\mathrm{III},1} = &-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\ &\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\ &\times\Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big]\end{aligned}\tag{10.1}$$

Group III Row V. The same, with $\frac{ig^4 f^{c'ca}}{4\pi^4}$ and its colour factor:

$$\begin{aligned}(\mathrm{B})_{\mathrm{V}} = &-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\ &\Big[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\,\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big]\end{aligned}\tag{10.2}$$

Group III Row III, second part. Here the bracket has six terms and an extra factor $\frac{1}{p^+(x-z)^2+k^+(x-w)^2} \simeq \frac{1}{p^+(x-z)^2}$ at large p^+. With $F(p^+)\simeq\frac{k^+}{p^++k^+}F^z$:

term of the bracket	integrand at $p^+\gg k^+$	$\int^{\vee-k^+} dp^+$
$\frac{\delta_{k'm}d_\perp p^+}{2(z-w)^2(p^++k^+)^2}[\dots]$	$\propto\frac{1}{p^{+3}}$	finite
$\frac{p^+\delta_{k'm}}{k^+(p^++k^+)}[\dots]$	$\propto\frac{1}{p^{+2}}$	finite
$\frac{\delta_{k'm}}{p^++k^+}[\dots]$	$\propto\frac{1}{p^{+3}}$	finite
$-[\frac{1}{p^+}]_+[\dots]$, $f(p^+)$	$\propto\frac{1}{p^{+3}}$	finite
$-[\frac{1}{p^+}]_+[\dots]$, $-f(0)$	$+\frac{f(0)}{p^+}$	cancels the $\log(\frac{\vee}{k^+}-1)$ of line 2 (Step 5)
$-\frac{p^+}{k^{+2}}[\dots]$	$-\frac{1}{k^+(x-z)^2}\,\frac{1}{p^++k^+}\,[\dots]F^z$	$\log\frac{\vee}{k^+}$: (B) (Step 6)
$-\frac{1}{k^+}[\dots]$	$\propto\frac{1}{p^{+2}}$	finite

The $-\frac{p^+}{k^{+2}}$ term, with $\frac{p^+}{p^+(x-z)^2+k^+(x-w)^2}\to\frac{1}{(x-z)^2}$ and $(x'-y')\to(x'-z)$, $U^{e'c'}(y')\to U^{e'c'}(z)$, gives

$$\begin{aligned}(\mathrm{B})_{\mathrm{III},2} = &-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}\\ &\times\left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2} - \frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]\\ &\Big[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\ &\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]\end{aligned}\tag{10.3}$$

Group III Row IV. Only four ρ's; nothing of this kind.

Group III Rows I, II. Their $\log\frac{\vee}{k^+}$ is not hidden: it is inside the logarithms, e.g. $\log\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{k^+(x-w)^2+\Lambda(x-z)^2} = \log\frac{\vee}{k^+} + \log\frac{(1-\frac{k^+}{\vee})(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2} + O(\Lambda)$, and $\log(\frac{\vee}{k^+}-1)$ there has no plus prescription to cancel it. It is taken out in `BFKL_rows.pdf` (blue).

Group II Row IV (the same in your variable ξ). With $\xi = \frac{k^+}{p^+ + k^+}$, $\frac{d\xi}{\xi(1-\xi)} = -\frac{dp^+}{p^+}$, your $\frac{\xi}{(1-\xi)_+}$ is your $[\frac{1}{p^+}]_+$, and its subtraction cancels the $\log(\frac{\vee}{k^+} - 1)$ of line 2, as in Step 5. The term that grows is $\frac{1-\xi}{\xi}$: with $\frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}$ and the phase $e^{-ik\cdot(y'-y)/\xi}$, the $-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}$ of your pole bracket has the transform $\int d^2\Delta\, e^{-iq\cdot\Delta}\log\Delta^2 = -\frac{4\pi}{q^2} = -\frac{4\pi\xi^2}{k^2}$. The integrand is then $\propto \frac{d\xi}{\xi}$, from $y' \to y$ (and $y, y' \to z$), which gives $\int_{k^+/\vee}^1 \frac{d\xi}{\xi} = \log\frac{\vee}{k^+}$ times your H_4 (line 2 of Group II Row IV in your `BFKL.tex`, which is correct). The constant parts of the bracket (pole, γ_E, -1) multiply functions that are smooth at $y' = y$; their transforms fall fast in $\frac{1}{\xi}$, so they give no logarithm. In `Rest_terms.pdf` the row therefore has $-\int_{k^+/\vee}^1 \frac{d\xi}{\xi}\, H_4$.

11 Why (B) is needed: the checks

- **BFKL⊗LO.** With the terms (B), the BFKL list equals the log $\frac{\vee}{k^+}$ list of `LL_terms.pdf` (4.4×10^{-17}), and its two-ρ terms are BFKL⊗LO (0 of 109 colour structures differ, `../final_check/render.py`). In the kernel-by-kernel comparison of `BFKL_final.pdf` all 67 structures agree with (B); without (B), 9 of the 67 differ. These are exactly the kernels in which the measured gluon is emitted by z: $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)$, $\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)$ and $K(w,w',z)K(x,x',z)$.
- **Three ρ's.** The three-ρ parts of the (B) terms cancel among themselves (Rows III and V: ∓ 3.6368, Row VI: 0, in the table of `BFKL_rows.pdf`, Sec. 6). The three-ρ test therefore passes with or without them (3.5×10^{-16} / 3.6×10^{-16}); (B) matters for the two-ρ terms.
- **The rest is finite only with $-$(B).** Toy model, Sec. 9: without $-$(B) the rest falls by $2.30\, F^z$ per decade of $\frac{\vee}{k^+}$.